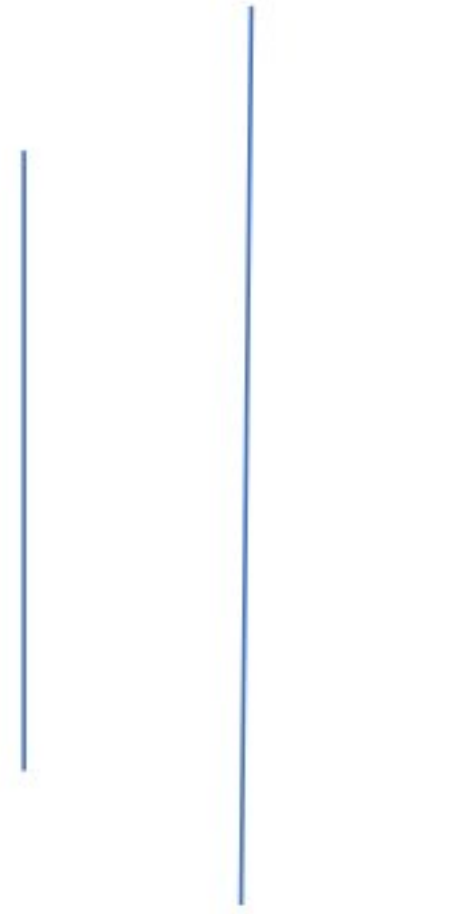


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A REPORT OF HYDRAULICS



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Experiment 1: Reynolds's experiment

Objective: To determine the type of flow in pipe.

Apparatus required:

- A tank containing water at constant head
- A small tank containing dye
- A horizontal glass tube with a bell mouthed entrance at one end and a regulating valve at another end.

Theory:

The Reynolds number (Re) is a dimensionless ratio of inertia force to viscous force and is used in determining the type of flow occurring: Laminar or Turbulent. The Reynolds number exist anywhere in the transition region is called the critical Reynolds number.

The objective of this experiment is to determine the range of Reynolds numbers by Reynolds apparatus over which the transition occurs. Given the tube (12mm), the Reynolds number can be calculated as;

$$Re = (\rho v D) / \mu$$

Where,

Re= reynolds number

ρ = density(kg/m³)

v= velocity of fluid (m/s)

D= hydraulic diameter (m)

μ =dynamic viscosity of fluid (Ns/m²)



Figure 1. Reynold number (Re) Apparatus

The hydraulic diameter (different than hydraulic radius) is calculated as:

$$D = (4 * \text{Area}) / \text{Wetted Perimeter}$$

Experimental Procedure:

1. Turn on the water, and partially open the discharge valve at the base of the apparatus.
2. Open and adjust the dye injector valve to obtain a fine filament of dye in the flow down the glass tube. If the dye is dispersed in the tube reduced to water flow rate by closing the discharge valve and adjusting the supply as necessary to maintain the constant head. A laminar flow condition should be achieved in which the filament of dye passes down the complete length of the tube without disturbance (Figure 2).
3. Record the temperature of the water using the thermometer, find the corresponding kinematic viscosity from a table.
4. Then measure the flow rate by timing the collection of a known quantity (volume) of water from discharge pipe. This will help you to determine the velocity of the water in the pipe.
5. Slowly increase the flow rate by opening the discharge valve until disturbances of the dye filament are noted (Figure 2). This can be regarded as the starting point of transition to turbulent flow. Increase the discharge as required to maintain constant head conditions. Do not miss to sketch the dye condition and measure the flow rate for each of your trials.
6. If necessary, increase the flow rate as described above until the disturbances increase such that the dye filament becomes rapidly diffused. Small eddies will be noted just above the point where the dye filament completely breaks down. This can be regarded as the onset of fully turbulent flow (Figure 2).
7. Now you should close the dye injector valve in order to finalize your experiment.

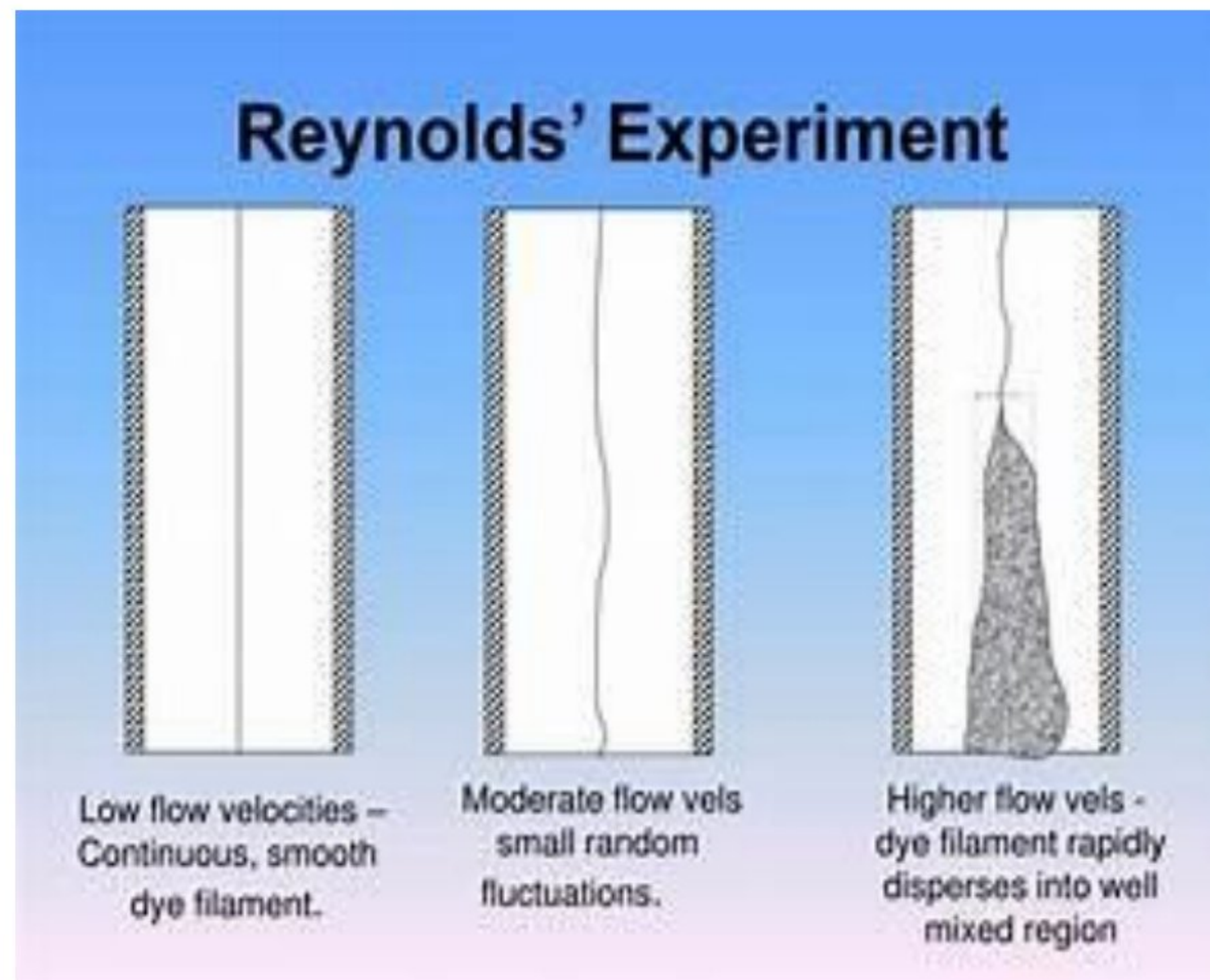


Figure 2. Dye sketches (Laminar through Turbulent)

Observation:

1. When the velocity of the flow was low, the dye filament in the glass tube was in the form of a straight line. This was a case of laminar flow.
2. With the increase in the velocity of flow, the dye filament was no longer a straight line, but it became wavy. This shows that the flow is no longer laminar. This was a transitional state.
3. With further increase in velocity, the wavy dye-filament broke up and finally diffused in water. This was the case of a turbulent flow.

Result :

1. The velocity at which the flow changes from laminar to turbulent for the case of a given fluid at a given temperature and in a given pipe is a critical velocity.
2. The occurrence of a laminar and turbulent flow was governed by the relative magnitude of inertia and viscous force.

Experiment no:2 flow through open sluice gate

Object: To investigate the operating characteristics of a sluice gate in open channels.

Scope: sluice gates are used In irrigation systems to control the flow rates. The study of characteristics of sluice gates provides the information associated for their hydraulic design.

Apparatus required:

- 1.open channel flow
2. stop watch

Theory:

- a) Flow through a rectangular orifice is

$$Q_{th} = A(2gH)^{0.5}$$

Where, Q_{th} = theoretical discharge

A = area of gate opening

H = head

g = acceleration due to gravity

- b) Coefficient of discharge $C_d = Q_a / Q_{th}$

- c) Actual discharge Q_a = volume/time



fig: flow through open channel

Procedure:

- 1.start the pump and allow water to flow through the flume with valve set at minimum opening.
2. set the gate opening at 25 mm.
3. operate the flow control valve to give a head 100mm in the tank and allow the condition of flow.

4. determine the flow rate by timing a known volume (m^3) passing through the flow meter at the bottom.
5. note the inlet head.
6. Now increase the gate opening to 50 mm and 75 mm while keeping the head at 100 mm by operating the flow control valve, measure the discharge at each step.
7. Repeat the procedure for heads in the tank of 150mm, 200mm, 250mm, 300mm, and 350mm for the sluice gate opening of 25mm, 50mm, and 75mm.

Calculations:

For head=100mm

Gate opening =25mm

Volume=0.1 mm^3

Time =102 sec

Width b=100mm

Area $A=25*100=2500\text{mm}^2$

a)actual discharge (Q_a) =volume/time

$$=0.1/102$$

$$=0.00098\text{m}^3/\text{s}$$

b)Theoretical discharge(Q_{th})= $A(2gH)^{0.5}$

$$=2500*10^{-6}*(2*9.81*0.1)^{0.5}$$

$$=0.003502$$

c)coefficient of discharge (C_d)= Q_a/Q_{th}

$$=0.28$$

Similarly, observation for other observations are tabulated as follows:

Observations :

Width of flume=100mm

No. of observation s	Head (mm)	Gate opening(mm)	Volume(mm ³)	Time(sec)	Discharge*10 ⁻³ m ³ /s)
1.	100	25 50 75	0.1	102 42.45 31.92	0.9804 2.36 3.133
2.	150	25 50 75	0.1	51 28.82 19.64	1.963 4.382 5.092
3.	200	25 50 75	0.1	33 18.19 13.54	3.03 5.498 7.386
4.	250	25 50 75	0.1	32 17.22 10.17	3.125 5.807 9.833
5.	300	25 50 75	0.1	30 14.28 10.64	3.33 7.003 9.398
6.	350	25 50 75	0.1	—	—

Based on the calculation, theoretical discharge and coefficient of discharge are calculated below:

Result table:

No. of obs.	Head (mm)	Gate opening	Actual discharge(10 ⁻³ m ³ /s)	Theoretical discharge (*10 ⁻³ m ³ /s)	Coefficient of discharge(Cd)
1.	100	25 50 75	0.9804 2.36 3.133	8.502 7.00 10.5	0.279 0.337 0.298
2.	150	25 50 75	1.963 4.382 5.092	4.28 8.557 12.8	0.459 0.512 0.398
3.	200	25 50	3.03 5.498	4.95 9.90	0.612 0.555

		75	7.386	14.8	0.499
4.	250	25	3.125	5.53	0.585
		50	5.807	11.07	0.525
		75	9.833	16.6	0.592
5.	300	25	3.33	6.06	0.55
		50	7.033	12.13	0.577
		75	9.398	17.57	0.535

Conclusion:

In this experiment, head and corresponding flow rate was observed. The actual and theoretical value of discharge were calculated for rectangular open channel and coefficient of discharge was found in the range of 0.279 to 0.612. the result clearly shows that the value of C_d decreases with increase in the sluice gate for same head. It is useful in flow of water in dams.

Experiment no. 3: open channel flow (hydraulic jump)

Objective: To compare the experimental value of depth before hydraulic jump to that calculated from theory and calculate the energy loss in hydraulic jump .

Scope: The formation of hydraulic jump is related with a sudden rise in water depth, large scale turbulent and dissipation of energy. It is employed at the foot of spillways and other hydraulic structures to dissipate energy for the protection of bed against scour. This experiment help to understand the features of hydraulic jump.

Apparatus required: a) open channel flow
b) stop watch

Theory:

Hydraulic jump is created by changing slope or placing a flow weir in the bed of the channel. It is mainly used to dissipate the energy and reduce velocity.

Here, $2q^2/g = y_1 y_2 (y_1 + y_2)$

Where, y_1 = depth before jump

y_2 = depth after jump

q = unit discharge

experiment

and, Energy loss $= (y_1 - y_2)^3 / 4 y_1 y_2$

Experimental procedure:

- (a) The pump was started and sluice gate was opened to about 25mm.
- (b) The flow rate was adjusted to give about 300mm bed above the sluice gate.
- (c) The adjustable weir was raised to form a hydraulic jump within the control portion of the flume.
- (d) The depth was noted before and after the jump.



Fig: hydraulic jump

(e) The flow rate and head were measured.

(f) The procedure was repeated for a head of 500mm above gate.

Observations:

Gate opening=26mm

Channel width=100mm

No. of obs.	Head in mm	Depth y1	Depth y2	Volume m ³	Time (sec)
1.	280	21	89	0.1	28.84
2.	345	28	100.4	0.1	27.63

Calculations:

(a) Discharge per unit width (q) = (volume/t)/d
 $= (0.1/28.84)/0.1$
 $= 0.0347 \text{ m}^2/\text{s}$

(b) We know, $2q^2/g = y_1 y_2 (y_1 + y_2)$
 For $q=0.0347$ and $y_2=0.089\text{m}$
 $2 \cdot 0.0347^2 / 9.81 = y_1 \cdot 0.089 (y_1 + 0.089)$
 Or, $y_1 = 24.3\text{mm}$.

Similarly, for $q=0.0362$ and $y_2=0.1004\text{mm}$
 $2 \cdot 0.0362^2 / 9.81 = y_1 \cdot 0.1004 (y_1 + 0.1004)$
 Or, $y_1 = 21.8\text{mm}$

Sample calculation

Energy loss (E_1) = $(y_2 - y_1)^3 / 4 y_1 y_2$

So,

$$E_{h1} = (0.089 - 0.0243)^3 / 4 \cdot 0.089 \cdot 0.0243$$

$$= 31.34 \text{ m head of water}$$

And, $E_{h2} = (0.1004 - 0.0218)^3 / 4 \cdot 0.1004 \cdot 0.0218$
 $= 55.46 \text{ m head of water}$

Result table:

No. of obs.	Discharge $Q(m^3/s)$	Unit discharge (m^2/s)	$Y_1(mm)$ theoretical	Experimental $E_1(mm)$	Theoretical $E_1(mm)$
1.	0.00347	0.0347	24.3	42.06	31.31
2.	0.00362	0.0362	21.8	54.54	55.46

Conclusion:

In this experiment, the value of depth before jump has been found and compared to the theoretical value. Similarly, energy loss was also seen and calculated which might be due to various practical factors. The phenomenon is useful in reducing velocity of flow by dissipating energy through jumps in open channel.